Report

Nara Avila Moraes

Physics Institute
IFUSP

2026 / 02 / 05

Summary

Tasks Status

Task	Status	Link
Criar toy model		Notebook
$\hookrightarrow$ Gerar dados sintéticos (correlacionados + ruído opcional)	✓	
$\hookrightarrow$ Ajustar GP (kernel, hiperparâmetros)	✓	
$\hookrightarrow$ Comparar: resíduos, log-likelihood	✓	
$\hookrightarrow$ Plotar a previsão	✓	

Tasks Status

Task	**Status**	
Analisar toy model		
↪ Construção do modelo GP	✓	
↪ Predição	✓	
↪ Resíduos	✓	
↪ Comparação entre incertezas experimentais e preditas	✓	
↪ Estudo das incertezas correlacionadas e não correlacionadas, experimentais e preditas	◑	
↪ Aprendizado do kernel	○	

Tasks Status

Task	**Status**	**Link**
Escrever projeto		Projeto
↪ Definir escopo e objetivos	✓	
↪ Esboçar introdução (motivação + observáveis)	✓	
↪ Listar método/dados (GP, Inferencia Bayesiana)	○	
↪ Listar objetivos	◑	
↪ Cronograma	✓	
↪ Selecionar bibliografia	✓	

Recent studies

Presentation	Status	Link
Exploring the universality of scaled particle spectra with Bayesian inference in heavy-ion collisions	✓	Link

Selected Paper

Estimation of reliability and accuracy of models of φ-sub-Gaussian process using generating functions of polynomial expansions.

Oleksandr Mokliachuk

Sun Pharma Advanced Research Company, Ltd. - Princeton, NJ,
e-mail: omokliachuk@gmail.com

February 5, 2026

Abstract. Stochastic processes are often represented through orthonormal series expansions, a framework originating in the classical works of Loève and Karhunen and widely used for simulation and numerical approximation. While truncation error in such expansions has been extensively studied, practical models frequently involve an additional source of error arising from the approximation of coefficient functions when closed-form expressions are unavailable. The combined effect of these two errors remains insufficiently addressed in the literature. Building on the author's earlier work on reliability and accuracy estimates for φ-sub-Gaussian processes, this paper extends the methodology to orthonormal polynomial systems that do not possess normalized generating functions in analytical form, including the Legendre, generalized Laguerre, and Gegenbauer families. New bounds are derived for models in $L_p(T)$ and $C([0,T])$ that simultaneously account for truncation and coefficient approximation. The resulting criteria provide practical guidance for selecting the number of series terms required to achieve prescribed levels of reliability and accuracy across a broader class of polynomial-based stochastic process models.

Keywords Models of stochastic processes, quasi-Banach spaces, Karhunen-Loève model, reliability and accuracy, sub-Gaussian random processes, Legendre polynomials, generalized Laguerre polynomials, Gegenbauer polynomials

AMS 2010 subject classifications. Primary: 60G07, 62M15, Secondary: 60G12, 60G15, 46E30

Figure: Estimation of reliability and accuracy of models of Gaussian